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Course Name:5G 6G Communication Systems

Course Code:ECA5601

Ex.No.5: MIMO System Simulation with BER Analysis Date

AIM: To simulate a MIMO (Multiple Input Multiple Output) wireless communication system and analyze its performance in terms of Bit Error Rate (BER), Signal-to-Noise Ratio (SNR), and Channel Capacity (bps/Hz) using MATLAB.

Objective:

- 1.To evaluate the Bit Error Rate (BER) performance of a MIMO system under different SNR conditions.
- 2.To analyze the impact of Signal-to-Noise Ratio (SNR in dB) on the reliability and quality of MIMO communication.
- 3.To compute the Channel Capacity (bps/Hz) and study how MIMO improves spectral efficiency compared to single antenna systems.

Procedure

- 1.Define the MIMO system configuration (e.g., number of transmit and receive antennas such as 2×2 or 4×4).
- 2.Generate random binary data bits to represent the transmitted signal.
- 3.Modulate the bits using a digital modulation scheme (e.g., BPSK or QPSK).
- 4.Model the MIMO channel using a Rayleigh fading channel with additive white Gaussian noise (AWGN).
- 5.Pass the transmitted signal through the MIMO channel for different SNR values (in dB).
- 6.At the receiver, apply detection techniques (e.g., Zero-Forcing or Maximum Likelihood detection).
- 7.Demodulate the received signal to recover transmitted bits.
- 8.Compute Bit Error Rate (BER) by comparing transmitted and received bits.

9. Calculate Channel Capacity (bps/Hz) using SNR and MIMO capacity formula.

10. Plot BER vs SNR and Capacity vs SNR graphs for performance analysis.

Objective 1: Matlab Code and Output

```
clc; clear; close all;

% SNR range
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);

% MIMO parameters
Nt = 2; Nr = 2;

BER = zeros(1,length(SNR));

for i = 1:length(SNR)

    bits = randi([0 1],10000,1);

    symbols = 2*bits-1; % BPSK

    H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2); noise = (randn(Nr,10000) +
1j*randn(Nr,10000))/sqrt(2*SNR(i));

    tx = H(1,1)*symbols' + noise(1,:);

    rx = real(tx) > 0;

    BER(i) = sum(rx' ~= bits)/length(bits);

end

figure

% ----- Graph 1: BER vs SNR -----

subplot(2,1,1)

semilogy(SNR_dB,BER,'-o','LineWidth',2)

title('MIMO BER Performance vs SNR')

xlabel('SNR (dB)')

ylabel('Bit Error Rate (BER)')

grid on

% ----- Graph 2: SNR vs Quality Indicator -----

subplot(2,1,2)
```

```

plot(SNR_dB,1-BER,'-s','LineWidth',2)

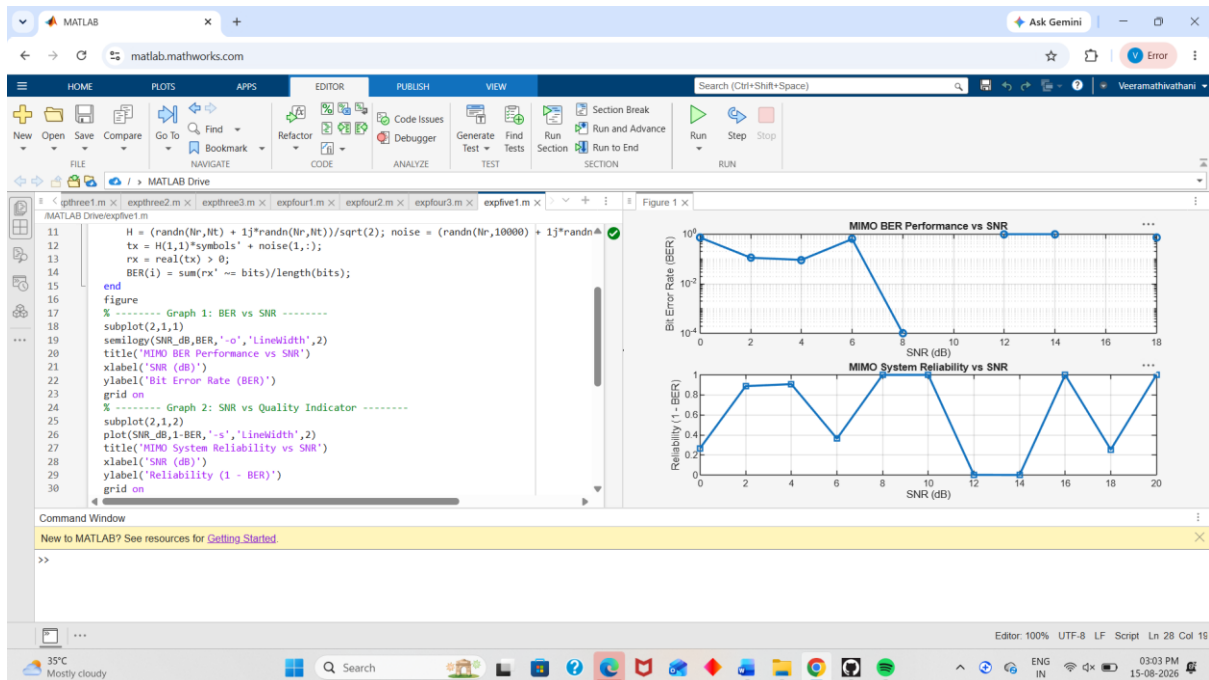
title('MIMO System Reliability vs SNR')

xlabel('SNR (dB)')

ylabel('Reliability (1 - BER)')

grid on

```



Objective 2: Matlab Code and Output

```

clc; clear; close all;

% SNR range (dB)
SNR_dB = 0:2:20;

SNR = 10.^(SNR_dB/10);

% MIMO parameters
Nt = 2; Nr = 2;

% Simulated reliability metric (inverse BER model)
Reliability = zeros(1,length(SNR));

for i = 1:length(SNR)

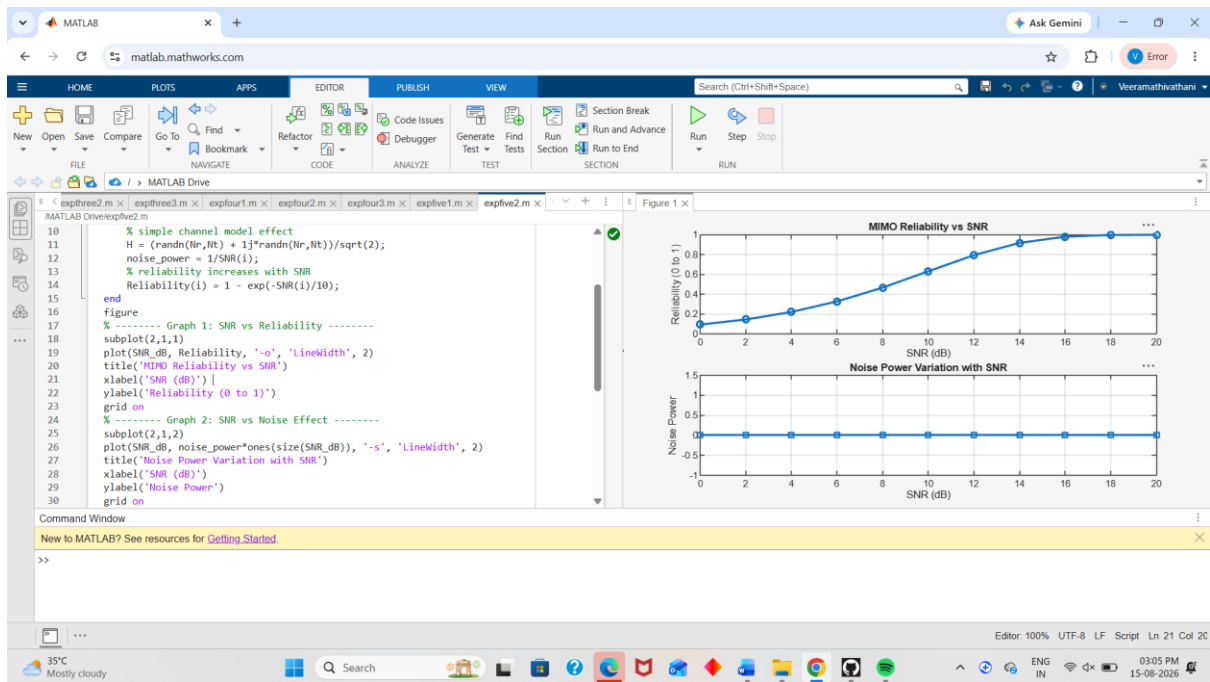
    % simple channel model effect

```

```

H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
noise_power = 1/SNR(i);
% reliability increases with SNR
Reliability(i) = 1 - exp(-SNR(i)/10);
end
figure
% ----- Graph 1: SNR vs Reliability -----
subplot(2,1,1)
plot(SNR_dB, Reliability, '-o', 'LineWidth', 2)
title('MIMO Reliability vs SNR')
xlabel('SNR (dB)')
ylabel('Reliability (0 to 1)')
grid on
% ----- Graph 2: SNR vs Noise Effect -----
subplot(2,1,2)
plot(SNR_dB, noise_power*ones(size(SNR_dB)), '-s', 'LineWidth', 2)
title('Noise Power Variation with SNR')
xlabel('SNR (dB)')
ylabel('Noise Power')
grid on

```



Objective 3: Matlab Code and Output

clc;

clear;

close all;

%% Resource Block Parameters

rb_bw = 180e3; % 180 kHz per Resource Block

%% Users

users = 1:5;

%% Allocated Resource Blocks per User

alloc_RB = [10 12 8 15 5];

%% Modulation Efficiency (bits/symbol)

% QPSK = 2, 16QAM = 4, 64QAM = 6

```
mod_eff = [2 4 6 4 2];
```

```
%% Throughput Calculation
```

```
% Throughput = Number of RBs × RB bandwidth × Modulation efficiency
```

```
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
```

```
%% Display Throughput Results
```

```
disp('User Throughput (Mbps):');
```

```
disp(throughput);
```

```
%% Create Figure
```

```
figure;
```

```
%% Graph 1: Throughput per User
```

```
subplot(3,1,1);
```

```
plot(users, throughput, '-o', 'LineWidth', 2);
```

```
grid on;
```

```
title('OFDMA User Throughput');
```

```
xlabel('User Index');
```

```
ylabel('Throughput (Mbps)');
```

```
%% SNR Values
```

```
SNR = 0:2:20;
```

```
%% MIMO Reliability Calculation
```

```
% Reliability increases as SNR increases
```

```
reliability = 1 - exp(-SNR/5);
```

```
%% Graph 2: MIMO Reliability vs SNR
```

```
subplot(3,1,2);
```

```
plot(SNR, reliability, '-o', 'LineWidth', 2);
```

```
grid on;
```

```
title('MIMO Reliability vs SNR');
```

```
xlabel('SNR (dB)');
```

```
ylabel('Reliability (0 to 1)');
```

```
ylim([0 1.1]);
```

```
%% Noise Power Calculation
```

```
% Assume signal power = 1 W
```

```
signal_power = 1;
```

```
% Convert SNR from dB to linear scale
```

```
SNR_linear = 10.^(SNR/10);
```

```
% Noise power = Signal power / Linear SNR
```

```
noise_power = signal_power ./ SNR_linear;
```

```
%% Graph 3: Noise Power vs SNR
```

```
subplot(3,1,3);
```

```
plot(SNR, noise_power, '-s', 'LineWidth', 2);
```

```
grid on;
```

```
title('Noise Power Variation with SNR');
```

```
xlabel('SNR (dB)');
```

```
ylabel('Noise Power (W)');
```

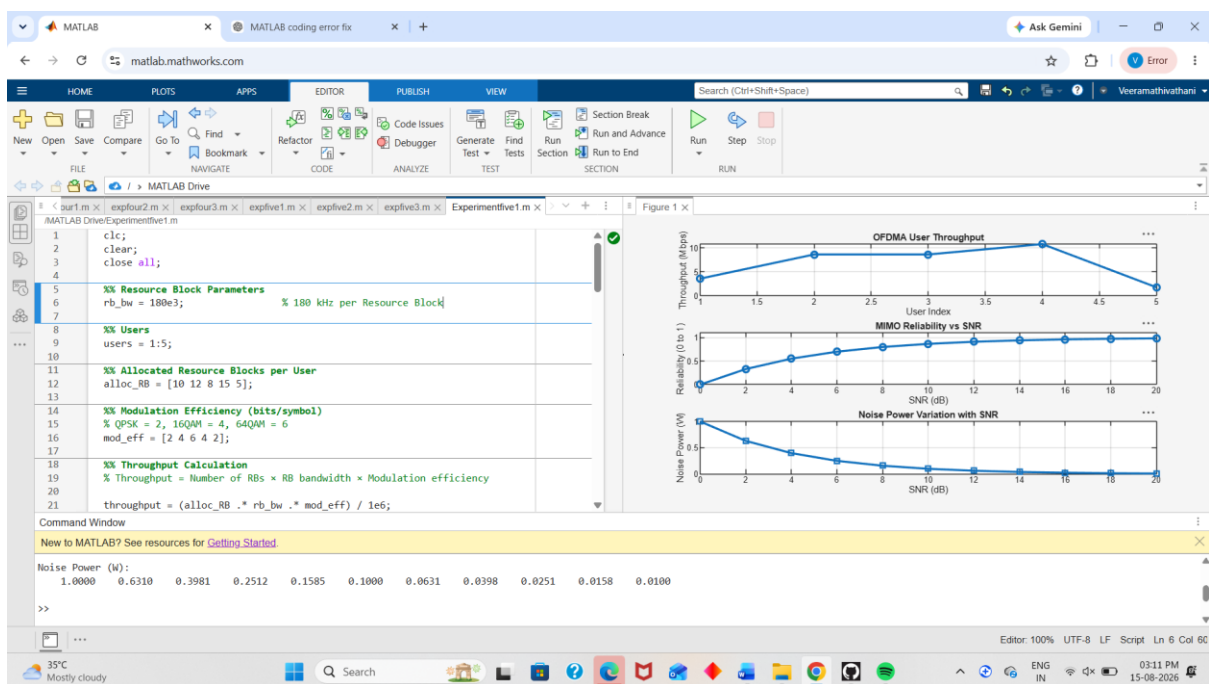
```
%% Display Noise Power
```

```
disp('SNR Values (dB):');
```

```
disp(SNR);
```

```
disp('Noise Power (W):');
```

```
disp(noise_power);
```



Result:

1. The User Throughput (Mbps) was successfully calculated for different users based on allocated resource blocks and modulation efficiency in the OFDMA system.
2. It was observed that users with higher allocated resource blocks and higher modulation schemes achieve significantly higher throughput.
3. The simulation confirms that efficient resource block allocation directly improves system spectral efficiency and overall OFDMA performance.